CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

The Automated Resource Management System is a web tool that helps organizations manage their resources.Its make easy for them to track information about resources.Its also helps them record the organization and tracking of information about the resources.This means that resources are used in a way that requires less less manual work to be done.Its helps the relevant people to make decisions by keeping records and generating reports on time.Its makes thing better.The Automated Resource Management System is different from methods that use papers and spreadsheets.Its maintains all information about resources in a database.It track the reviews of the resources if they are available and create dashboards.These dashboards display summaries and reports about the resources and the status of the inventory.The main goal of the Automated Resource Management System is to provide organizations with a way to manage resources that is cost-effective.The Automated Resource Management System facilitates the allocation of resources, make records more accurate,aids in decision-making.It does this through data management, and report generation in a systematic way.Its helps organizations manage their resources in a way.

The main objective of the Automated Resource Management System are the following:

- To automate the process of recording and storing information about resources.
- To manage the allocation and availability of resources in a systematic way.
- To monitor the status of resource inventory.
- To create dashboards and reports that analyze data.
- To reduce errors in record-keeping.
- To help managers make decisions based on data.

The Automated Resource Management System was created using established principles of software engineering. It has a user interface,a way to automatically generate reports.The system has components that work together to record the management,monitoring,and evaluation of resources

Figure 4.1 illustrates the overall workflow of the developed system.

(Insert System Workflow Diagram Here)

The workflow begins with user login.After they log in,authorized users can access the system based on their roles.The authorized personnel manage information about the resources,allocate the resources,update inventory information,and monitor the usage of the resources.The Automated Resource Management System maintains all transactions in the database. It generates reports that are displayed on dashboards and graphical summaries. These reports help the relevant individuals to assess resource availability,identify usage trends,and make management decisions. The Automated Resource Management System is a tool for organizations.

4.3 System Development Methodology

The development of the Automated Resource Management System used the Agile Software Development Methodology.The Agile methodology is about building things in iterations,collaborating all the time, and regularly reviewing the progress.The Agile methodology was chosen because it allows the development team to quickly respond to changing requirements. It also incorporates feedback from stakeholders at each stage of development. The Agile methodology divides the project into iterations or sprints. At the end of each sprint,the development team built,tested,reviewed, and refined the parts of the system.This approach minimized the risks.The quality of the software was improved.It also helped fix the system issues.The development process has the following stages:

The development process consisted of the following phases:

Requirements Gathering

This stage is about collecting information from users.This was done by communicating with them, observing them, and studying documents.Consulted the relevant authorities and staff.They helped identify the challenges that exist in resource management.These challenges include resource management in record tracking,delayed reporting,and inaccurate inventory tracking.The collected information served as the basis for determining the requirements and functional details of the Automated Resource Management System. It is designed to address these challenges and make resource management easier.By implementing this system,organization can streamline their processes,enhance accuracy, and improved overall efficiency. The integration of real-time data analytics will also enable better decision-making and resource allocation.

System Planning

This stage focuses on identifying the objectives,scope,and necessary functions of the project.The researchers gathered data through interviews,observations,and discussions within the organization to better understand the current management processes. At this stage,the project timeline,resource allocation, and the feasibility of implementing the proposed system were also examined.The goal is to ensure that the system effectively addresses the problems experienced in the manual management of resources.

System Design

In the design phase,the overall structure of the system was carefully prepared.This includes designing the database,user interface,process flow, and system architecture.The wireframes,Entity Relationship Diagrams(ERD), and Unified Modeling Language(UML) diagrams were created to provide a clear plan for the system’s implementation. The researchers also designed user-friendly interfaces to ensure that employees and administrators can easily navigate the application with minimal training.

Software Development

The development stage involves coding and integrating various modules of the Automated Resource Management System.The application was developed using modern programming languages and web technologies suitable for database-driven applications.Each module has been built individually and continuously tested before integration to ensure smooth operation and maintain software quality.

Testing

To ensure the reliability and accuracy of the system,various levels of software testing were conducted.Unit testing verified each individual module, while integration testing ensured all modules communicated correctly.System testing examined the overall performance ,security, and usability of the application.Finally User Acceptance Testing (UAT)was conducted with selected users to determine if the system met their expectations and operational requirements.The feedback collected at this stage was used to improve the final version of the application.

Deployment

After the successful testing ,the Automated Resource Management System was deployed in the target environment.The deployment process includes configuring the database server,installing the application,creating user accounts, and transferring existing resource records when necessary.The researchers also provided orientation and basic training to the users to help then become familiar with the feature and capabilities of the system.

(*Insert the narratives of your deployment here..coordinate with your UI/x)*

Maintenance

The maintenance phase ensures that the system continues to operate efficiently after deployment. At this stage,software updates,bug fixes, database optimization,and feature enhancements are carried out as needed.Users feedback is continuous.

Figure 4.2 presents the Agile Software Development Life Cycle adopted in the study.

(Insert Agile Methodology Diagram Here or any SLDC model)

4.4 System Architecture

The Automated Resource Management System(ARMS)uses a **Three-Tier Client-Server Architecture**, which separates the application into three main layers:the Presentation Layer,Application Layer,and Data Layer.This architecture improves scalability,maintainability,and security by assigning specific responsibilities to each layer.It also allows for easier system upgrades and simplifies troubleshooting whenever changes are needed.

Figure 4.3 illustrates the architecture of the developed system.

(Insert System Architecture Diagram Here)

The architecture consists of the following components:

Presentation Layer

It serves as the communication interface between the users and the application. It provides responsive and user-friendly pages that allow users to perform various resource management tasks conveniently using desktop computers,laptops,tablets, or smartphones through a web browser.

The main components of the interface include:

- Login Page
- Dashboard

- Resource Management
- Resource Allocation
- Reservation Module
- Availability Tracking
- User Management
- Reports
- Notifications
- Profile Settings

The interface is designed to be simple, organized, and easy to navigate,allowing users to perform their responsibilities with minimal effort

Application Layer

It contains the business logic responsible for processing the requests submitted by users. It invalidates the input data,interacts with the database,and performs all resource management operations.

Its main responsibilities include:

- User Authentication
- Resource Registration
- Reservation Processing
- Resource Allocation
- Availability Monitoring
- Inventory Updates
- Report Generation
- Notification Processing
- Audit Trail Recording

This layer serves as a bridge between the user interface and the database,ensuring that all information is processed accurately while enforcing system policies and security measures.

Data Layer

It stores all of the organization’s information in a centralized relational database.It is responsible for maintaining data integrity.reducing redundancy, and supporting efficient record retrieval.

The stored information includes:

- User Accounts
- Departments
- Resources
- Resource Categories
- Reservations
- Allocations
- Borrowing History
- Maintenance Records
- Notifications
- System Logs
- Reports

The database is structured using normalization techniques to improve consistency and optimize system performance.The architecture also ensures secure communication between the application server and the database server while preventing unauthorized access to the organization's confidential information.

4.5 Hardware and Software Requirements

The successful implementation of the Automated Resource Management System depends on having the right combination of hardware and software resources.These requirements are essential to ensure that the system runs smoothly,responds efficiently to user requests,and provides reliable performance in daily use. With the appropriate setup,users can easily manage resources,monitor activities real-time, and generate accurate reports without unnecessary delays or technical issue.Meeting these specifications also helps improve the stability of the system,making it a more seamless experience for its users.

Additionally, investing in regular maintenance and updates can further enhance system performance and security.By staying proactive in managing these resources,organizations can adapt to evolving demands and ensure long-term success.

4.5.1 Hardware Requirements

Table 4.1 presents the recommended hardware specifications.

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	256 GB SSD	512 GB SSD
Monitor	1366 × 768 Resolution	Full HD Display
Internet Connection	10 Mbps	50 Mbps
Mobile Device	Android 10	Android 13 or higher

The recommended hardware configuration provides sufficient processing capability for executing analytical operations and rendering graphical dashboards without significant delays.

4.5.2 Software Requirements

Table 4.2 presents the software requirements.

Software	Purpose
Windows 11 / Ubuntu Linux	Operating System
Visual Studio Code	Source Code Editor
MySQL	Database Management System

PHP / Node.js Backend Development

ReactJS Frontend Development

Google Chrome Web Browser

Git Version Control

XAMPP Local Development
Server

The selection of these software tools was based on their stability, community support, compatibility, and widespread adoption in modern web application development.

4.6 Database Design

The database serves as the foundation of the Automated Resource Management System by organizing and storing user information ,resource records,reservations,allocations,maintenance schedule and system reports.

A well-designed database reduces data redundancy,maintains coin consistency, and ensures accurate information retrieval during daily operations.The researchers adopted a relational database model using mySQL and app normalization techniques to improve efficiency and maintain relationships in the related tables.

Figure4.4 shows the Entity Relationship Diagram(ERD)of the proposed system

(Insert the ERD Diagram) of the database of the database include:

- Users
- Departments
- Resources
- Resource Categories
- Reservations
- Allocations
- Maintenance Log
- Notifications
- Audit logs
- Reports

These entities work together to support the overall resource functionality of the Automated Resource Management System, ensuring accurate resource tracking ,efficient allocation, and reliable reporting. The integration of these entities allows for a leveraging advanced query optimization and indexing strategies,the database can handle complex transactions and provide real-time insights into resource utilization and availability.

Field Description

user_i d   Name Login

fullna me

Username

userna me

passwo rd   Encrypted
Primary Key Password

Complete

role User Access
Level

Customers Table

This table contains customer demographic information used for behavioral analysis.

Field Description

customer _id	Primary Key
fullname	Customer Name
age	Customer Age
gender	Customer Gender
address	Customer Address

Products Table

This table stores product information.

Field	Description
product_id	Primary Key
product_name	Product Name
category_id	Product Category
price	Selling Price
stocks	Available Inventory

Transactions Table

This table records sales transactions.

Field	Description
transaction_id	Primary Key
customer_id	Foreign Key
transaction_date	Date Purchased
total_amount	Total Purchase

Transaction Details Table

This table records the individual items included in each transaction.

Field Description

detail_id Primary Key
Foreign Key
transactio
n_id

product_id Foreign Key

quantity Quantity
Purchased

subtotal Item Total

Analytics Table

The analytics table stores computed insights generated by the application. Depending on the implementation, this table may contain aggregated metrics or references to dynamically computed results.

Field Description

analytics_id Primary Key

customer_id Foreign Key

preferred_cate gory Number of Purchases

purchase_frequ ency
Average Spending
average_spendi ng
Most Frequently
Purchased Category

generated_date Date of Analysis

The relationships among these tables allow the application to trace customer purchases, identify recurring buying behaviors, calculate product demand, and generate comprehensive analytical reports. Proper indexing and foreign key constraints help maintain referential integrity and optimize query performance, ensuring that the system can efficiently support both operational transactions and data analytics.

CHAPTER 5

(5 to 10 pages..)

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter provides a comprehensive conclusion to the development of the TOPBuy App. It synthesizes the project's results, assesses whether the research objectives were achieved, and offers recommendations for future enhancements and implementation strategies to further maximize the system's utility for small and medium-sized enterprises (SMEs).

5.2 Summary of Findings

The development and evaluation of the TOPBuy App yielded significant results aligned with the research objectives:

- The system successfully automated the recording and storage of sales transactions, eliminating manual data entry errors. • Analytical modules effectively processed historical data to identify purchasing patterns, allowing business owners to predict inventory needs with higher accuracy.
- Visual dashboards provided clear, actionable insights into product popularity and customer segments, directly supporting data-driven decision-making.
- User acceptance testing confirmed that the interface is intuitive, reducing the learning curve for business owners and staff.

5.3 Conclusion

The TOPBuy App has proven to be a viable and effective solution for SMEs seeking to optimize their business intelligence. By transforming raw transaction data into behavioral insights, the system empowers owners to make informed strategic decisions regarding inventory management, promotional activities, and customer relationship management. The project successfully addressed the need for an affordable, responsive, and data-centric tool, demonstrating that sophisticated analytics can be made accessible to small-scale enterprises. The research questions were answered affirmatively, confirming that the system enhances operational efficiency and competitive advantage.

5.4 Recommendations

To ensure the continued relevance and efficacy of the TOPBuy App, the following recommendations are proposed:

System Improvements

- Integrate automated notification alerts for low-stock items and expiring products to further improve inventory management.
- Enhance security protocols by implementing two-factor authentication to better protect sensitive sales and customer data.
- Develop advanced export features allowing reports to be generated in multiple formats (e.g., PDF, CSV, Excel) for easier sharing and integration with accounting software.

Future Research Directions

- Explore the integration of machine learning algorithms to enable predictive analytics, such as forecasting future sales trends based on seasonal data.
- Conduct a longitudi\nal study to analyze the long-term impact of the system on business growth and profitability for SMEs.
- Investigate the feasibility of a mobile-native application to provide offline functionality for businesses with intermittent internet connectivity.

Implementation Strategies

- Business owners should conduct training sessions for staff to ensure consistent data entry habits, which are critical for accurate analytics.
- SMEs are encouraged to perform regular data backups and system maintenance to ensure security and prevent data loss.

5.5 Impact of the Study

The TOPBuy App contributes to the IT field by demonstrating the practical application of data analytics in a resource-constrained environment. By providing SMEs with accessible business intelligence, this study bridges the gap between complex data processing technologies and everyday business operations. While the project faced limitations—such as the need for comprehensive historical data to generate accurate behavioral patterns—the results underscore the significant potential for technology to level the playing field for small businesses in an increasingly data-driven market.

5.6 Summary

In summary, the TOPBuy App represents a successful integration of software engineering principles and data analytics. It serves as a testament to the power of tailored technology solutions in addressing real-world business challenges. This research provides a solid foundation for future developers to build upon, contributing to the broader goal of digital transformation among small and medium-sized enterprises.